TANGO: mapping flood damage from space

Space Track challenge report. Case study: Trishuli flood, Rasuwa, Nepal, 26 August 2026. Repository: https://github.com/bhoopesh282008/Tango

> **14 October 2026.** The pipeline has run end to end on real Sentinel-1 scenes for five areas in Nepal and India, and its output for the case study has been compared with the EMSR927 reference (section 5). Live dashboard: https://bhoopesh282008.github.io/Tango/. In short: **almost all of what it maps lies inside the reference corridor, but it finds only a small part of the damage**, between 3% and 35% of the reference area depending on the reach. The flood figures below are therefore lower bounds.

1. What the system does

TANGO takes an area and a flood date and answers three questions for rescue teams: where the flood hit, what was damaged, and who is cut off. It has two parts.

- **A pipeline** (Python, `pipeline/`) that finds a before/after Sentinel-1 pair, calibrates and terrain-corrects it, maps flood water and debris, overlays pre-event OpenStreetMap, and works out which settlements have lost their road connection to a hospital, and whether they can still reach a town. One command runs it: `python run.py --bbox W,S,E,N --event YYYY-MM-DD`.
- **A dashboard** (React) that shows the result on a map with statistics, a rescue priority ranking, a one-page situation report and a copilot that answers four questions in English and Nepali.

There is no server. The pipeline writes static files and the dashboard reads them.

2. Data and rules

Data	Use	Source
Sentinel-1 GRD (VV, VH)	Before/after radar scenes	Copernicus Data Space
Copernicus DEM GLO-30	Terrain correction, slope, flood path	AWS open data
OpenStreetMap, 27 July 2026	Buildings, roads, bridges, health facilities, places	ohsome API, with Overpass API as fallback
Kuro Siwo	Training the flood model	Hugging Face, CC BY 4.0
Copernicus EMS EMSR927	Checking only	not used as input

No published damage map and no post-event OpenStreetMap edit is used as an input. The OpenStreetMap snapshot is taken 30 days before the event (never later than 27 July 2026) and the fetch refuses any date on or after it. Datasets outside the list above (modelled population, river flow) are not fetched unless asked for with `--context`, which is off by default; nothing in the results reads them.

One departure from the brief. The ohsome API returned HTTP 403 on its geometry extraction endpoints while its metadata and count endpoints answered normally. The pipeline still asks ohsome first, and when refused takes the same snapshot date from the Overpass API, which serves the same OpenStreetMap history. The output records which service supplied the data. All OpenStreetMap figures in this report came through Overpass.

3. Method

Scene pair. The pipeline searches the Copernicus catalogue 20 days either side of the event and keeps only pairs on the same relative orbit and direction, 12 days apart, with the event between them. Scenes from different tracks view the terrain from different angles and cannot be compared pixel by pixel. Among valid pairs it prefers one whose scenes cover at least 97% of the area, then the earliest image after the event. For Trishuli three tracks have a valid pair; it selects track 85 ascending, 16 and 28 August 2026, whose post-event

image is two days after the flood. Track 19 would give one ten days after, and track 121 covers only 43% of the area.

Preprocessing, written from scratch without ESA SNAP. Digital numbers are calibrated to sigma0 with the product's tables, speckle is reduced with a Lee filter, and the image is terrain-corrected by the Range-Doppler method onto a 10 m UTM grid (DEM heights converted from geoid to ellipsoid first). Layover and shadow are masked and both scenes share the grid, so they are aligned by construction. On the real scenes the result reproduces the product's own geolocation grid to 0.03 pixels and correlates 0.80 with brightness simulated from the DEM, best at zero shift. One correction came only from real data: the product's slant-to-ground records are not exact at their stated times, and using them as stamped misplaced pixels by up to 57 m.

Flood map. A flood fills valley floors, so change counts as flood only on ground within 30 m above and 600 m of a river (from pre-event OpenStreetMap, or from DEM-derived channels where no river is mapped). There, a drop of more than 3 dB is water or wet sediment and a rise of more than 3 dB is debris. Elsewhere only strong change (4.5 dB) over at least a hectare is kept, as "uncertain". The first version required water to be darker than -18 dB; on the real scenes only 5% of darkened river pixels are that dark, because a river tens of metres wide in a gorge never looks like open water at 10 m. The 30 m and 600 m limits are our assumptions; they were not tuned against any reference map. The alternative method uses the trained model for water (section 4) and the same rule for debris.

Steadier baseline (the default). A fixed threshold on one "before" image cannot tell ordinary change from flood: run on two scenes that are both from before the event, the rule marks as much as 24% to 30% of the mapped area in the two weeks before the flood and 94% to 127% in the monsoon onset. So "before" is the median of three images on the same track (the "before" scene and the two 12 days apart before it), and a change counts only if it also exceeds 3 robust standard deviations of that pixel's variation across the three; the 3 was fixed in advance, not tuned on a map. On the same pre-event pairs it marks 9% to 13% of its own area. On the Bhote Koshi in Sindhupalchok (not the Rasuwa Bhote Koshi of the case study; no flood there on the event date is confirmed), an August and a February date give the same area under the single-image rule (0.19 and 0.18 km^2) and different areas under the baseline (0.16 and 0.05 km^2); the seasons differ too, so this shows how the baseline behaves, not that it finds a flood. It needs three scenes on one track; with fewer it warns and falls back (`--baseline-images 1` forces the single image).

Optical evidence. With `--optical` the pipeline adds Sentinel-2 composites per pixel (the latest clear look before the event and the earliest after, with cloud, shadow and snow rejected by the scene classification layer). New water is a water index (MNDWI) turning positive; debris is vegetation (NDVI) disappearing. Optical never removes a radar detection and follows the same valley-floor rule; it raises confidence where both agree, fills radar blind spots at lower confidence, and marks change that only it sees as "uncertain".

Damage. Buildings, roads and bridges that intersect a water or debris zone are flagged. This is an overlap, not an inspection, so the dashboard labels them "in flood zone" and not "destroyed". Most rural roads have no name in OpenStreetMap, so an unnamed road is listed by its kind and nearest settlement.

Cut-off settlements. Roads become a graph, and each settlement is placed on it once, before the cut. The headline count closes every road that touches a water or debris zone; a firmer count closes only the stretches inside a zone. A settlement is cut off when it had a road route to a hospital before the event and has none afterwards. A settlement with no mapped road before the event is reported as "access unknown", so that gaps in the map are not counted as flood damage. Whether the settlement can still reach a town (a place OpenStreetMap tags as town or city) is reported beside it, because the brief asks for the nearest town or hospital; a run with no town mapped, such as Trishuli, says nothing rather than guess.

Flood path (bonus). From any point, the path is traced on the DEM by expanding outward and always taking the lowest cell on the frontier. Unlike plain steepest descent, this climbs out of pits and DEM noise. The dashboard runs it in the browser when the user taps the map, and lists the settlements within 500 m of the path in downstream order.

4. AI component: flood segmentation model

A small U-Net (7.8 million weights) labels each pixel as no water, permanent water or flood from four bands: post-event VV and VH, pre-event VV and VH.

Training. We trained on a 10 GB sample of Kuro Siwo read as chunks spread evenly over every shard, so that every flood event is represented: 4,240 training patches, of which 709 patches from three whole events were held out for validation. 40 epochs took about 35 minutes on a laptop GPU (RTX 3050, 6 GB).

Results on the Kuro Siwo test sample, flood class.

Test patches	Count	Model IoU	Model F1	Threshold IoU	Threshold F1
Floods absent from training	312	0.46	0.63	0.27	0.43
Floods also present in training	858	0.69	0.82	0.45	0.62
All	1,170	0.67	0.80	0.44	0.61

How to read this. 11 of the 18 test activations also occur in the training sample, as other areas of the same flood. The first row is therefore the honest "never seen" figure. On those seven floods the IoU ranges from 0.16 to 0.82, and three are at about 0.20 or below. The model beats the threshold rule in every row, mostly by finding more of the flood (recall 0.74 against 0.29 on unseen floods) with more false alarms (precision 0.55 against 0.81). A stricter test retrains with every test event left out (320 patches): on the 12 events outside the 1111xxx series, whose numbering interleaves with training events, it scores 0.55 against 0.34.

Himalayan scenes. Almost none of the Kuro Siwo sample is mountainous: only 36 test patches have more than 300 m of relief, all from floods also present in training. On the real Trishuli scenes, which the model has never seen, it marks 0.53 km² of water on valley floors against 1.62 km² for the single-image threshold rule (0.43 km² common to both; the model cannot use the baseline). It fires only where the surface became very dark (median -15 dB), at the Betrawati confluence and parts of the channel below, and finds nothing in the upper gorge near Rasuwagadhi. Against EMSR927 (section 5) it does worse than the threshold rule in all three areas: 2% to 14% of the reference area against 3% to 22%, at similar precision (0.90 to 0.97). The gain on Kuro Siwo does not carry over to this valley.

5. Results for the Trishuli area

Flood and damage map. Scenes: Sentinel-1D, track 85 ascending, 16 and 28 August 2026 (baseline from 23 July and 4 August). A first run on a new area takes 4 to 16 minutes on a laptop, mostly reading scenes or waiting for the free OpenStreetMap service; a repeat from saved rasters takes under a minute.

	Default (baseline)	Single image	Model for water
Water or wet sediment	1.39 km²	1.62 km²	0.53 km²
Debris	0.68 km²	0.89 km²	0.89 km²
Buildings in a flood zone	563 of 85,186	672	154
Road segments in a flood zone	91 of 2,059 (8.4 km inside a zone)	121 (10.8 km)	56
Bridges in a flood zone	12 of 192	14	7
Settlements cut off (only flooded stretches closed)	12 of 149 (7)	23, earlier cut-off code	15 (11)

The valley floor is 2.8% of the area and 7% of the scene is lost to layover and shadow. Within 100 m of rivers the share of pixels that changed by more than 3 dB is about twice the share elsewhere, so the corridor signal is real, but part of it may be ordinary change in a monsoon river between two dates. The cut-off count rests on a few marginal road sections: with only the flooded stretches closed, 7 of the 12 stay cut off.

Comparison with EMSR927. The reference was used only after the runs, with nothing tuned on it. `python validate.py <run> cache/reference_emsr927` reproduces every figure here. The four reference areas with products are covered by two runs on the same scene pair: the main run above, and a second run to the south-west (84.9 to 85.25 E, 27.75 to 28.02 N) made with the same code and settings as a test on another area. The second run maps 4.7 km² (669 of 124,059 buildings, 159 road segments, 29 of 176 bridges, 14 of 116 settlements cut off, 13 of them also without a road to a town). The reference records the event as a mass movement and outlines the whole debris corridor by photo-interpretation of 0.3 m to 1 m optical images of 27 to 31 August. Parts the analysts marked "not analysed" are excluded.

Area (run)	Reference	Ours (water + debris)	IoU	Precision	Recall	Model map: IoU / recall
01 Syapru Besi (main)	1.11 km²	0.03 km²	0.03	0.88	0.03	0.02 / 0.02
02 Timure (main)	1.29 km²	0.05 km²	0.04	0.99	0.04	0.02 / 0.02
03 Bidur, northern 63% (main)	4.09 km²	0.95 km²	0.21	0.93	0.22	0.14 / 0.14
03 Bidur, whole (second)	5.89 km²	1.36 km²	0.21	0.94	0.22	not run
05 Phosretar (second)	4.79 km²	1.73 km²	0.35	0.97	0.35	not run

Damage (default map)	Main run: areas 01, 02, 03 north	Second run: areas 03, 05
Reference buildings destroyed, damaged or possibly damaged; those in our zones	2,538; 541 (21%)	2,952; 756 (26%)
Buildings we flagged; those next to a reference building	332; 314 (95%)	456; 407 (89%)
Reference road segments destroyed or damaged; those our zones touch	170; 76 (45%)	333; 152 (46%)
Reference road segments with no visible damage; those our zones touch	197; 3 (2%)	522; 25 (5%)
Reference bridges destroyed or damaged; in our OpenStreetMap; flagged	20; 12; 6	18; 10; 7 (Bidur only)

The model column is from the single-image main run. The baseline costs little: recall falls slightly (0.23 to 0.22 at Bidur), precision is unchanged.

Reading. The map can be trusted where it says "flood" and cannot be trusted where it says nothing. Recall is poor because of what the radar pair shows, not mainly because of hidden slopes: inside the reference corridor under 7% of pixels are in layover or shadow, but 57% to 81% changed by less than 3 dB between the two dates. Only 15% of the corridor in the two gorge areas, 29% at Bidur and 43% at Phosretar crosses the 3 dB threshold in either direction, which caps what any rule of this kind can find; recall rises down the valley as the floor widens. Fresh debris over a gravel river bed or bare ground looks much the same to C-band radar at 10 m, while it is obvious in a 0.3 m optical image. Roads do better than buildings because a long segment needs only one detected crossing.

Optical evidence on real scenes. With `--optical` (on the single-image maps), Sentinel-2 gave a clear view before and after for 31% and 55% of the two areas. It confirms 0.07 km² of radar detections and fills about 0.01 km² of radar blind spots. Its "uncertain" patches, where radar saw no change, carry real signal on valley floors (counting them, recall against the reference would be 0.33 to 0.65, at precision 0.70 to 0.96) and are mostly thin cloud and haze elsewhere (58 to 82 km² per run). We have not promoted them to a flood class, because that rule would have been chosen after looking at the reference.

Other results on real data:

Item	Result
OpenStreetMap, 27 July 2026	85,186 buildings, 2,059 road segments (1,583 km), 192 bridges, 16 health facilities, 149 named settlements
Settlements with a road route to a hospital before the event	84 of 149; the other 65 are "access unknown" (87% of buildings are within 300 m of a mapped road)
Settlements with a population in OpenStreetMap	1 of 149
Flood path from Rasuwagadhi	59.9 km down the Trishuli; median 13 m from the OpenStreetMap river line, 90% within 42 m

With population recorded for one settlement, the dashboard reports mapped buildings instead of people and does not convert one into the other.

Two more Himalayan events. To test areas, years and orbit tracks the system had not seen, two glacier floods in India were run with the same code. Chamoli (Uttarakhand, 7 February 2021) straddles two image frames: read as one, the radar covered 84% of it and the baseline could not be built; with the frames joined it covers 100% (track 56, 29 January and 10 February) and the run maps 0.22 km^2 of water and debris, no building in a flood zone and no settlement cut off (30 of 61 have no mapped road). The South Lhonak glacial-lake flood down the Teesta valley (Sikkim, 4 October 2023; track 12, 25 September and 7 October) maps 2.9 km^2 of water and debris, 172 of 8,829 buildings in a flood zone and 1 of 31 settlements cut off. Neither was compared with a reference map, so they show that the system runs on a new event, not how well it maps it.

6. Dashboard and situation report

The dashboard shows the flood zones, buildings, roads, settlements and infrastructure on a map with a choice of basemaps; headline figures with breakdowns; a rescue priority ranking of cut-off settlements; and export to CSV, GeoJSON and a printable report. A before/after viewer shows the radar pictures with the mapped flood outlined on them, and a route tool plans the road from anywhere to a settlement or hospital: one route that avoids every flagged road section and the fastest, with the distance and the stretches inside a flood zone, and live GPS navigation. Times use assumed speeds by road class, since nothing in the data gives speeds. The copilot answers "where did the flood hit", "what infrastructure is damaged", "which settlements are cut off" and "which should be rescued first" in English and Nepali, and assembles the four into a situation report.

Every number comes from the maps. One function derives all statistics from the loaded layers; the cards, the ranking, the exports and the copilot read from it. The copilot is not a language model: its answers are templates filled with those figures, so it cannot invent one. Typed questions are matched by keyword to the four topics, and anything else gets a plain "cannot answer".

Missing is not zero. Where the data lacks something the dashboard says so: unknown road access, population not recorded, power lines not assessed. The priority score uses only the factors recorded for every cut-off settlement, so a gap in the data cannot lift one settlement above another.

The live dashboard opens on the case study; its area selector also holds the model's map, the second reach and the rehearsals. A copy built without run data shows an illustrative demo dataset, labelled "Demo data".

7. Limitations

- **No early warning.** Sentinel-1 returns to the same track every 12 days. The system maps a flood after the next pass; it cannot warn of a glacier collapse or a flood in progress. For Trishuli the earliest usable post-event scene is two days after the event.
- **Optical helps little as built.** Cloud left 31% and 55% of the two areas with a clear view on both sides of the event, and optical patches off the valley floor are mostly false (section 5).

- **Most of the damage is missed.** Against EMSR927 the map finds 3% to 35% of the affected area and 21% to 26% of the affected buildings (section 5). The flooded area and the damage counts are lower bounds. The cut-off count is not: missed flood can hide a cut-off, and closing whole roads can make one. Rescue planning must not treat an unmarked area as safe.
- **False alarms remain.** Even the steadier rule marks 9% to 13% of its own mapped area between two images from before the flood (section 3), so part of every map is ordinary change, not the flood. The EMSR927 precision cannot show this: its reference is a wide corridor that contains the river.
- **The comparison is partial.** It covers one event and the four reference areas that have products. The 30 m, 600 m and 3 dB limits were not tuned on the reference, and we have not tried to raise recall, because fitting them to EMSR927 would make this comparison meaningless.
- **Frames and swaths.** A Sentinel-1 pass is cut into frames along the orbit, and an area near a cut lies in two (a rehearsal on the 2021 Chamoli flood: 84% and 16%). The pipeline reads and joins every frame of the pass; where the pass still misses part of the area, the dashboard states the share covered and the rest is not mapped. The join has had one real test so far (Chamoli: 84% with one frame, 100% joined).
- **Steep terrain.** Radar cannot see slopes in layover or shadow (7% of the scene). In the upper gorge near Rasuwagadhi, where this flood began, almost nothing is detected.
- **Valley floors only.** Flood classes are limited to ground near a mapped river, using limits we chose. A debris flow down an unmapped gully is at best marked "uncertain".
- **The model does worse than the thresholds here** and has no debris class. Debris always comes from fixed thresholds, which can also fire on wet soil, crops, snow and unrelated landslides.
- **Damage is an overlap.** A building in a flood zone is counted whether or not it was damaged; one destroyed outside the mapped zone is not. At 10 m, single houses are not resolved.
- **OpenStreetMap is incomplete.** Footpaths and small settlements may be missing; 65 of 149 settlements have no mapped road, and Trishuli maps no town, so "can it still reach a town" has no answer there. Building counts stand in for population, and not every building is a home. Multipolygon buildings are skipped.
- **Road cuts are coarse.** The headline count closes every road that touches a flood zone from end to end; closing only the flooded stretches leaves 7 of Trishuli's 12 cut off. Both are shown. Most mapped "hospitals" are unnamed and probably health posts, which makes the cut-off test easier to pass than it should be.
- **Simplifications.** The flood path is a drainage line (direction, not depth, width, timing or reach). Preprocessing uses the product's orbit data, not precise orbit files, has no radiometric terrain flattening and applies only the range part of the noise tables.
- **Nepali text** was written without review by a native speaker.
- **Not an operational tool.** Map tiles come from free public services with no availability guarantee, and nothing here has been checked in the field.

No images of people affected by the disaster are used anywhere in the system or this report.

8. Attribution

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Bountos, N. I., Sdraka, M., Zavras, A., Karavias, A., Karasante, I., Herekakis, T., Thanasou, A., Michail, D., Papoutsis, I. (2024). Kuro Siwo: 33 billion m² under the water. A global multi-temporal satellite dataset for rapid flood mapping. Advances in Neural Information Processing Systems 37.